Combinatorics of the critical loci of Deep Linear Networks

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Abstract

We stratify the space of critical points of the loss landscape of deep linear networks for the quadratic loss. The stratification provides a combinatorial description of the minima and saddle points of a deep linear network. The stratification is parameterized by specific geometric invariants, the cyclic rank patterns, which are the ranks of partial products of the layers of a DLN.

Contents

1 Introduction **1**
1.1 Motivation . . . 1
1.2 Setup and main theorem . . . 2
1.3 Related work . . . 4

2 A stratification of the cyclic variety induces a stratification of the critical loci **4**

3 The cyclic rank patterns as geometric invariants of deep linear networks **8**
3.1 The cyclic Nakayama algebra . . . 8
3.2 From multiplicity patterns to rank patterns . . . 13

4 Stratification of the critical loci with cyclic rank patterns **14**

5 Discussion **18**
5.1 Future work . . . 19

6 Acknowledgements **20**

7 Appendix **21**
7.1 Inversion Formula . . . 21

1 Introduction

1.1 Motivation

The motivation for this combinatorial description is that it can provide a coarse-grained description of the loss landscape that distinguishes between different strata of degenerate solutions. Because the parameter-function map is highly degenerate, i.e. it has a whole group of symmetries, so is the loss, and there are infinitely many parameters that map to the same DLN. Singular learning theory (SLT) Watanabe [2009] shows that model degeneracies are important and related to generalization in the context of Bayesian statistics. We also know that deep neural networks are highly degenerate

and that their internal symmetries should influence generalization Wei et al. [2022]. The work of Lehalleur and Rimányi [2024] shows that the codimension of the largest strata in the stratification of the global minima is exactly related to the learning coefficient of SLT, which informally measures the amount of degeneracy in a degenerate statistical model. However, frontier deep learning models such as large language models are not typically trained to convergence. To better understand the solutions selected by a training process before convergence, we want to have better models of such solutions. In this regime, it is important to model the effect of critical points that are not global minima and can also govern the training dynamics. An example of such a phenomenon is the saddle-to-saddle dynamics in deep linear networks Jacot et al. [2018], Saxe et al. [2019]. In the saddle-to-saddle regime of deep linear networks, gradient flow performs a low-rank approximation of a teacher matrix, where a transition between saddles corresponds to increasing the rank of the student by one. A combinatorial description of all the critical points enables us to coarse-grain the loss landscape into a graph where each node corresponds to a stratum and the dynamics of the training process can be modeled as transitions between these strata. In summary, our combinatorial description can provide a global description of the training dynamics.

1.2 Setup and main theorem

The central result of this paper is that the algebraic variety of critical points of the loss landscape of a deep linear network admits a stratification parameterized by a finite set of geometric invariants given by rank patterns. These rank patterns are the ranks of products made of weight matrices and a projection of the input-output covariance matrix.

Definition 1.1 (Deep Linear Network)**.** An L-layer Deep Linear Network (DLN) is a function:

$$f : \mathbb{R}^{d_0} \times \prod_{l=0}^{L-1} \mathbb{R}^{d_l \times d_{l+1}} \to \mathbb{R}^{d_L}$$
$$(x, \theta = (W_1, \dots, W_L)) \mapsto \mu(\theta)x = W_L \cdots W_1 x$$

The dimension d_0 is the dimension of the input data, and the dimensions d_l, $1 \leq l \leq L$, are the widths of the layers. A Deep Linear Network is a multilinear function in parameter space $\Omega = \prod_{l=0}^{L-1} \mathbb{R}^{d_l \times d_{l+1}}$, which is linear in each weight matrix W_l and in the data. It is linear in function space $\mathcal{F} := \mathrm{Mat}_{d_L \times d_0}$ over $\mu(\theta) := W := W_L \cdots W_1$, where μ is the product map. For the input-output data $(X, Y) \sim P$, we have the input and input-output covariance matrices $\Sigma_X = \mathbb{E}[XX^\top]$ and $\Sigma_{YX} = \mathbb{E}[YX^\top]$. Throughout the paper, without loss of generality, we assume that the inputs are **whitened**, i.e. $\Sigma_X = I_{d_0}$, and we work with the quadratic population loss

$$L(\theta) := \|M - \mu(\theta)\|_F^2,$$

where $\|\cdot\|_F$ is the Frobenius norm and $M := \Sigma_{YX}\Sigma_X^{-1} = \Sigma_{YX}$ is the teacher matrix, which is the ordinary least-squares regressor.

The teacher matrix M admits the following SVD:

$$M = U\Lambda V^\top$$

with $\Lambda = \mathrm{diag}(s_1, \dots, s_{d_L}, 0_{d_0 - d_L})$ and $s_1 > s_2 > \cdots > s_{d_L} > 0$ (full-rank and nondegenerate data). We **assume a realizable setting**, i.e. $d_l \geq d_0 \geq d_L$ for l running over the hidden layers of the DLN, and thus the global minimum is $W^\star = M = \Sigma_{YX}$ with corresponding loss zero. Let S be a subset indexing singular vectors of M. We define P_S to be the projector onto these singular vectors and P_Q to be the projector onto the singular vectors orthogonal to those indexed by S. The projectors P_S and P_Q are orthogonal to each other. Recall Proposition 1 of Achour et al. [2024]:

Proposition 1.2. Let θ be a critical point. Then there exists a subset S of singular directions of M such that

$$\mu(\theta) = P_S W^\star = P_S M$$

Under the whitened input assumptions, this means that $\mu(\theta) = P_S \Sigma_{YX}$. We denote $W_S := P_S \Sigma_{YX}$ and $W_S^\dagger := \Sigma_{XY} P_Q$. We have $W_S^\dagger W_S = 0$ by orthogonality of P_S and P_Q, and since $M^\top = \Sigma_{XY}$, we also have $W_S W_S^\dagger = 0$, as P_S and P_Q project onto orthogonal subsets of left singular vectors of M.

We define some algebraic varieties that will be important for deriving the main result. The cyclic variety Γ_d is defined by a set of vanishing polynomial cyclic conditions related to the critical loci. This variety will then be stratified, and the critical loci will inherit a stratification.

Definition 1.3 (Determinantal, cyclic, and critical loci)**.** Define

$$\Sigma_d^r := \{\theta \in \Omega : \operatorname{rank}(\mu(\theta)) = r\} \quad \textit{Determinantal variety},$$

$$\Gamma_d := \{(\theta, W_0) \in \Omega \times \mathrm{Mat}_{d_0 \times d_L} \mid W_{<\ell} W_0 W_{>\ell} = 0 \text{ for all } 1 \le \ell \le L\}, \quad \textit{Cyclic variety}$$

Let $\mathcal{F}^t := \mathrm{Mat}_{d_0 \times d_L}$, and let $\pi : \Gamma_d \to \mathcal{F} \times \mathcal{F}^t$ be the map $\pi(\theta, W_0) := (\mu(\theta), W_0)$. Define the determinantal cyclic variety:

$$\Gamma_d^r := \Gamma_d \cap (\Sigma_d^r \times \mathrm{Mat}_{d_0 \times d_L})$$

Let S be a subset of left singular vectors of M, and let P_S be the projector onto the space spanned by these left singular vectors. Define the S-critical locus:

$$\mathcal{C}_d^S = \{\theta \in \Omega \mid \nabla_\theta L(\theta) = 0,\ \mu(\theta) = P_S W^\star\}.$$

In Proposition 2.1, we show that the S-critical locus is a fiber of π:

$$\mathcal{C}_d^S \cong \Gamma_d^S := \pi^{-1}(W_S, W_S^\dagger) \subset \Gamma_d^r.$$

Definition 1.4 (Symmetry group)**.** For the dimension vector $d = (d_0, \ldots, d_L)$, define the symmetry group:

$$G_d = \prod_{i=0}^{L} GL_{d_i}$$

Consider the action of G_d:

$$g.(\theta, W_0) = (g_L W_L g_{L-1}^{-1}, \ldots, g_1 W_1 g_0^{-1}, g_0 W_0 g_L^{-1})$$

and note that we have a well-defined action of G_d on the cyclic variety by G_d-invariance of the equations defining the cyclic variety. The variety Γ_d^r also inherits an action of G_d from the action of G_d on $\Sigma_d^r \times \mathrm{Mat}_{d_0 \times d_L}$, since rank is invariant under the action of GL_d. A consequence of the action is that we can stratify the variety Γ_d^r into orbits (strata) O_ρ. In this paper we will show that the relevant geometric invariants ρ parameterizing the orbits are the cyclic rank patterns.

Definition 1.5. For $\phi = (\theta, W_0) \in \Gamma_d$, we define its *cyclic rank pattern* to be the array

$$\rho(\phi) = (r_{ij}(\phi))_{0 \le i,j \le L} \quad \text{where} \quad \rho_{ij}(\phi) := \operatorname{rank}\big(W_j \cdots W_i\big), \quad \rho_{ii}(\phi) = d_i, \quad \rho_{i+1,i}(\phi) = \operatorname{rank}\big(W_i\big),$$

with indices read cyclically.

Theorem 1.6. Define the strata $\mathcal{O}_\rho^S := \mathcal{O}_\rho \cap \Gamma_d^S$, where $\mathcal{O}_\rho$ is a G_d-orbit of the cyclic variety Γ_d^r. We also identify $\mathcal{C}_d^S$ with Γ_d^S. Assuming that the data are nondegenerate and the input covariance matrix is invertible, the locus of critical points of rank $r < r_{\max}$ admits the following stratification, parameterized by subsets S of singular vectors of the teacher with r elements and admissible cyclic rank patterns $\rho \in R_d^{\mathrm{orb}}$:

$$\mathcal{C}_d^r = \bigsqcup_{|S|=r} \mathcal{C}_d^S = \bigsqcup_{|S|=r} \bigsqcup_{\substack{\rho \in R_d^{\mathrm{orb}} \\ \mathrm{rank}(\rho)=\mathrm{rank}(\Sigma_{XY} P_Q)}} \mathcal{O}_\rho^S$$

1.3 Related work

A first important result is that the loss landscape of a deep linear network has no nonglobal local minima Baldi and Hornik [1989], Kawaguchi [2016]. A second-order analysis has also extended these results to classify the saddle points of the DLN loss landscape, giving conditions on the ranks of partial products of the weights that classify them into strict and non-strict saddle points Achour et al. [2024]. Furthermore, when the DLN has all its layer widths strictly larger than the input and output dimensions, all global minima are connected Trager et al. [2019]. DLNs have whole groups of symmetries, and their study could be simplified by coarse-graining their internal symmetries. This is the perspective taken in Lehalleur and Rimányi [2024], which essentially shows that the global minima admit a stratification giving a combinatorial and hierarchical description of the minima via *rank patterns*. This work is an extension of the previous paper, where we derive a combinatorial description of all the critical points, including saddle points. The important geometric invariants are the rank patterns, rather than the cyclic rank patterns, which arise from cyclic vanishing equations describing the space of critical points beyond the constraint that the parameter-function map, namely the product map, maps to a particular critical value.

2 A stratification of the cyclic variety induces a stratification of the critical loci

In this section we explain how the orbit stratification of the cyclic variety induces an orbit stratification of each fixed-S critical locus. All closures are relative Zariski closures.

Fix a subset S of left singular vectors of M. We have $W_S \in \mathcal{F} = \mathrm{Mat}_{d_L \times d_0}$ and $W_S^\dagger \in \mathcal{F}^t = \mathrm{Mat}_{d_0 \times d_L}$. Recall that

$$\pi : \Gamma_d \longrightarrow \mathcal{F} \times \mathcal{F}^t, \qquad \pi(\theta, W_0) := (\mu(\theta), W_0).$$

We write

$$\Gamma_d^S := \pi^{-1}(W_S, W_S^\dagger) = \{(\theta, W_0) \in \Gamma_d \mid \mu(\theta) = W_S,\ W_0 = W_S^\dagger\}.$$

If $\mathrm{rank}(W_S) = r$, then $\Gamma_d^S \subseteq \Gamma_d^r$.

Proposition 2.1 (The S-critical locus as a fiber)**.** The map

$$\Theta_S : \mathcal{C}_d^S \longrightarrow \Gamma_d^S, \qquad \Theta_S(\theta) := (\theta, W_S^\dagger),$$

is an isomorphism of varieties, with inverse given by projection onto the θ-coordinates. In particular,

$$\mathcal{C}_d^S \cong \pi^{-1}(W_S, W_S^\dagger).$$

Proof. Let $\theta \in \Omega$ satisfy $\mu(\theta) = W_S$. Then

$$(P_S W^\star)^\top - \Sigma_{XY} = -\Sigma_{XY} P_Q = -W_S^\dagger.$$

Hence, for every $1 \le \ell \le L$,

$$\big(\nabla_{W_\ell} \mathcal{L}(\theta)\big)^\top = -2 W_{<\ell} W_S^\dagger W_{>\ell}.$$

Therefore the gradient equations are equivalent, under the constraint $\mu(\theta) = W_S = P_S W^\star$, to the cyclic equations

$$W_{<\ell} W_S^\dagger W_{>\ell} = 0 \qquad \text{for every } 1 \le \ell \le L.$$

These are precisely the equations saying that

$$(\theta, W_S^\dagger) \in \Gamma_d$$

together with $\pi(\theta, W_S^\dagger) = (W_S, W_S^\dagger)$. Thus

$$\theta \in \mathcal{C}_d^S \quad \Longleftrightarrow \quad (\theta, W_S^\dagger) \in \Gamma_d^S.$$

The maps $\theta \mapsto (\theta, W_S^\dagger)$ and $(\theta, W_S^\dagger) \mapsto \theta$ are regular and inverse to one another. □

We now use the symmetry action. The group

$$G_d = \prod_{i=0}^{L} GL_{d_i}$$

acts on Γ_d by

$$g \cdot (\theta, W_0) = (g_L W_L g_{L-1}^{-1}, \dots, g_1 W_1 g_0^{-1}, g_0 W_0 g_L^{-1}).$$

The map π is equivariant in the following sense:

$$\pi(g \cdot (\theta, W_0)) = (g_L \mu(\theta) g_0^{-1},\, g_0 W_0 g_L^{-1}).$$

Define the stabilizer of the fixed pair $(W_S, W_S^\dagger)$ by

$$K_S := \left\{ g \in G_d \;\middle|\; g_L W_S g_0^{-1} = W_S,\ g_0 W_S^\dagger g_L^{-1} = W_S^\dagger \right\}.$$

Then K_S acts on the fiber Γ_d^S.

Assume that the rank-r cyclic variety has a finite G_d-orbit stratification by cyclic rank patterns:

$$\Gamma_d^r = \bigsqcup_{\rho \in R_d^r} O_\rho,$$

where R_d^r denotes the finite set of cyclic rank patterns occurring in Γ_d^r. We assume moreover[1] that the ambient cyclic orbit closures are governed by the coordinatewise rank order:

$$\overline{O_\rho}^{\Gamma_d^r} = \bigsqcup_{\substack{\rho' \in R_d^r \\ \rho' \le_{\mathrm{rk}} \rho}} O_{\rho'}.$$

[1] Both assumptions are proven later in the paper.

Here

$$\rho' \leq_{\mathrm{rk}} \rho \quad \Longleftrightarrow \quad \rho'_{ij} \leq \rho_{ij} \quad \text{for every cyclic rank coordinate } (i,j).$$

For each $\rho \in R_d^r$, define

$$\mathcal{O}_\rho^S := O_\rho \cap \Gamma_d^S = O_\rho \cap \pi^{-1}(W_S, W_S^\dagger).$$

Let

$$R_d^S := \{\rho \in R_d^r \mid \mathcal{O}_\rho^S \neq \varnothing\}.$$

Proposition 2.2 (Induced K_S-orbit stratification of the S-critical locus)**.** For every $\rho \in R_d^S$, the stratum

$$\mathcal{O}_\rho^S = O_\rho \cap \pi^{-1}(W_S, W_S^\dagger)$$

is a single K_S-orbit. Consequently,

$$\Gamma_d^S = \bigsqcup_{\rho \in R_d^S} \mathcal{O}_\rho^S$$

is a finite decomposition into smooth locally closed K_S-orbits. Via the identification $\mathcal{C}_d^S \cong \Gamma_d^S$ from Proposition 2.1, this gives a stratification of $\mathcal{C}_d^S$. Indeed, define the fixed-S closure relation on R_d^S by

$$\rho' \leq_S \rho \quad \Longleftrightarrow \quad \mathcal{O}_{\rho'}^S \subseteq \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S}.$$

Equivalently, since the strata are K_S-orbits,

$$\rho' \leq_S \rho \quad \Longleftrightarrow \quad \mathcal{O}_{\rho'}^S \cap \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \neq \varnothing.$$

Then the closure of each stratum is the union of the strata below it for this relation:

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} = \bigsqcup_{\substack{\rho' \in R_d^S \\ \rho' \leq_S \rho}} \mathcal{O}_{\rho'}^S.$$

Equivalently, if

$$\rho' <_S \rho \quad \Longleftrightarrow \quad \rho' \leq_S \rho \text{ and } \rho' \neq \rho,$$

then the frontier satisfies

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \setminus \mathcal{O}_\rho^S = \bigsqcup_{\substack{\rho' \in R_d^S \\ \rho' <_S \rho}} \mathcal{O}_{\rho'}^S.$$

Moreover, the fixed-S closure order refines the ambient coordinatewise rank order:

$$\rho' \leq_S \rho \quad \Longrightarrow \quad \rho' \leq_{\mathrm{rk}} \rho.$$

Thus

$$\{\rho' \in R_d^S \mid \rho' \leq_S \rho\} \subseteq \{\rho' \in R_d^S \mid \rho' \leq_{\mathrm{rk}} \rho\}.$$

The inclusion may be strict.

Proof. First we show that each nonempty $\mathcal{O}_\rho^S$ is a single K_S-orbit. Let

$$x, y \in \mathcal{O}_\rho^S.$$

Since x and y both lie in the G_d-orbit O_ρ, there exists $g \in G_d$ such that

$$y = g \cdot x.$$

Because both x and y lie in the fiber Γ_d^S, we have

$$\pi(x) = (W_S, W_S^\dagger), \qquad \pi(y) = (W_S, W_S^\dagger).$$

Using the equivariance of π,

$$(W_S, W_S^\dagger) = \pi(y) = \pi(g \cdot x) = (g_L W_S g_0^{-1},\, g_0 W_S^\dagger g_L^{-1}).$$

Hence $g \in K_S$. Therefore $y \in K_S \cdot x$. Conversely, K_S preserves both the G_d-orbit O_ρ and the fiber Γ_d^S, so

$$K_S \cdot x \subseteq O_\rho \cap \Gamma_d^S.$$

Thus

$$\mathcal{O}_\rho^S = K_S \cdot x.$$

Since algebraic group orbits are smooth and locally closed, each $\mathcal{O}_\rho^S$ is smooth and locally closed. The decomposition

$$\Gamma_d^S = \bigsqcup_{\rho \in R_d^S} \mathcal{O}_\rho^S$$

follows by intersecting the disjoint orbit decomposition of Γ_d^r with the fiber Γ_d^S and discarding the empty intersections.

We now prove the frontier condition. Fix $\rho \in R_d^S$ and set

$$Z_\rho := \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S}.$$

Because $\mathcal{O}_\rho^S$ is K_S-stable, its closure Z_ρ is also K_S-stable. Suppose

$$Z_\rho \cap \mathcal{O}_{\rho'}^S \neq \varnothing.$$

Choose $x \in Z_\rho \cap \mathcal{O}_{\rho'}^S$. Since $\mathcal{O}_{\rho'}^S$ is a K_S-orbit,

$$\mathcal{O}_{\rho'}^S = K_S \cdot x.$$

Since Z_ρ is K_S-stable,

$$K_S \cdot x \subseteq Z_\rho.$$

Therefore

$$\mathcal{O}_{\rho'}^S \subseteq Z_\rho.$$

This proves the frontier condition. The closure equality follows because the strata $\mathcal{O}_{\rho'}^S$ form a finite disjoint decomposition of Γ_d^S.

It remains to compare $\leq_S$ with the coordinatewise rank order. Since

$$\mathcal{O}_\rho^S \subseteq O_\rho \qquad \text{and} \qquad \Gamma_d^S \subseteq \Gamma_d^r,$$

we have

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \subseteq \Gamma_d^S \cap \overline{O_\rho}^{\Gamma_d^r}.$$

Using the ambient cyclic closure formula,

$$\overline{O_\rho}^{\Gamma_d^r} = \bigsqcup_{\substack{\rho'' \in R_d^r \\ \rho'' \leq_{\mathrm{rk}} \rho}} O_{\rho''},$$

we obtain

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \subseteq \Gamma_d^S \cap \bigsqcup_{\substack{\rho'' \in R_d^r \\ \rho'' \leq_{\mathrm{rk}} \rho}} O_{\rho''} = \bigsqcup_{\substack{\rho'' \in R_d^S \\ \rho'' \leq_{\mathrm{rk}} \rho}} \mathcal{O}_{\rho''}^S.$$

Therefore, if $\rho' \leq_S \rho$, then

$$\mathcal{O}_{\rho'}^S \subseteq \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S}$$

forces

$$\rho' \leq_{\mathrm{rk}} \rho.$$

This proves

$$\leq_S \subseteq \leq_{\mathrm{rk}}$$

on the set R_d^S. □

Remark 2.3. The relation $\leq_S$ is the actual frontier relation on the fixed-S critical locus. The coordinatewise order $\leq_{\mathrm{rk}}$ is only an ambient necessary condition for specialization. Thus one should not replace $\leq_S$ by $\leq_{\mathrm{rk}}$ in the closure formula.

Indeed, even if

$$\rho' \leq_{\mathrm{rk}} \rho,$$

the intersection

$$O_{\rho'} \cap \pi^{-1}(W_S, W_S^\dagger)$$

may be empty. In that case there is no K_S-orbit $\mathcal{O}_{\rho'}^S$ in the fixed-S critical locus.

In the following sections we will show how to stratify Γ_d^r using quiver representation theory.

3 The cyclic rank patterns as geometric invariants of deep linear networks

3.1 The cyclic Nakayama algebra

We first recall some basic notions of quiver representation theory. A quiver is a directed graph $Q = (I, E)$ with vertices I and arrows E, where multiple arrows and loops are allowed. A quiver representation assigns a vector space $V_i \cong \mathbb{R}^{d_i}$ to each vertex and a linear map $\phi_i : V_{i-1} \to V_i$, with dimension vector $d = (d_i)_{i=0,\ldots,L}$. We denote by $\mathrm{Rep}_d(Q, k)$ the space of quiver representations of dimension d over a field k, which will be $\mathbb{R}$ or $\mathbb{C}$ for us. It is functorially equivalent to the category of kQ-modules, where kQ is the path algebra of the quiver Q generated by vertices and composable arrows.

Throughout this subsection, put $n = L + 1$, and take all indices modulo n. Unless otherwise stated, all A-modules are finite-dimensional left A-modules.

Let $Q = C_{L+1}$ be the cyclic quiver with vertex set

$$Q_0 = \{0, \dots, L\}$$

and arrows

$$a_l : l-1 \longrightarrow l, \qquad l \in \{0, \dots, L\},$$

where $a_0 : L \to 0$. We read products of arrows from right to left. For $1 \le l \le L$, define the length-L path

$$p_l := a_{l-1} \cdots a_1 a_0 a_L \cdots a_{l+1}.$$

Thus p_l is the unique path of length L starting at l, ending at $l-1$, and passing through the closing arrow a_0.

Let

$$I = \langle p_1, \dots, p_L \rangle$$

and define the bound path algebra

$$A := kQ/I.$$

Equivalently, an A-module with dimension vector $d = (d_0, \dots, d_L)$ is a representation of Q on vector spaces $V_i \simeq k^{d_i}$, with linear maps

$$\phi_l : V_{l-1} \longrightarrow V_l,$$

satisfying the relations

$$\phi_{l-1} \circ \cdots \circ \phi_1 \circ \phi_0 \circ \phi_L \circ \cdots \circ \phi_{l+1} = 0, \qquad 1 \le l \le L.$$

We write $\mathrm{Rep}_d(A)$ for the corresponding representation variety.

Proposition 3.1. For $k = \mathbb{R}$, the cyclic variety Γ_d is identified with $\mathrm{Rep}_d(A)$ by

$$W_l = \phi_l, \qquad V_i \simeq \mathbb{R}^{d_i}.$$

Proof. Under this identification, the defining equations of Γ_d are exactly the relations generated by the paths $p_1, \dots, p_L$. Hence $\Gamma_d \cong \mathrm{Rep}_d(A)$. □

Let $R \subset kQ$ be the arrow ideal, i.e. the ideal generated by all arrows. Every path of length $L+1$ contains one of the paths $p_1, \dots, p_L$ as a subpath. Therefore

$$R^{L+1} \subset I.$$

It follows that A is finite-dimensional. Moreover,

$$J(A) = R/I,$$

so $J(A)$ is spanned by the nonzero paths of positive length, and

$$A/J(A) \cong \bigoplus_{i=0}^{L} k e_i.$$

In particular, the simple A-modules are

$$S_i := P_i/JP_i, \qquad P_i := Ae_i.$$

Definition 3.2. A finite-length module U is called *uniserial* if its submodules are totally ordered by inclusion. Equivalently, U has a unique composition series.

A finite-dimensional algebra is called a Nakayama algebra if its indecomposable projective and indecomposable injective modules are uniserial.

Proposition 3.3. The algebra $A = kQ/I$ is a finite-dimensional Nakayama algebra. Its Kupisch series is

$$(c_0, \dots, c_L) = (L+1, L, \dots, L).$$

Proof. We first describe the indecomposable projectives. With our convention for path multiplication, $P_i = Ae_i$ is spanned by the nonzero paths starting at i. Let

$$q_{i,0} := e_i, \qquad q_{i,\ell} := a_{i+\ell} \cdots a_{i+1}$$

be the unique path of length ℓ starting at i. The path $q_{i,\ell}$ is nonzero precisely for

$$0 \le \ell < c_i,$$

where

$$c_0 = L+1, \qquad c_i = L \quad \text{for } 1 \le i \le L.$$

Indeed, the path starting at 0 of length L does not pass through a_0, so it is nonzero, whereas the path starting at 0 of length $L+1$ vanishes. If $i \neq 0$, then the path of length L starting at i is exactly the relation p_i, so it vanishes.

Thus

$$P_i = \langle q_{i,0}, q_{i,1}, \dots, q_{i,c_i-1} \rangle_k,$$

and

$$J^s P_i = \langle q_{i,s}, q_{i,s+1}, \dots, q_{i,c_i-1} \rangle_k, \qquad 0 \le s \le c_i.$$

In particular,

$$P_i \supset JP_i \supset J^2 P_i \supset \cdots \supset J^{c_i} P_i = 0.$$

Moreover,

$$J^s P_i / J^{s+1} P_i \cong S_{i+s}.$$

Hence this chain is a composition series.

It remains to check that it is the unique composition series. Let $N \subset P_i$ be a nonzero submodule. Since the idempotents e_v act as projections and each space $e_v P_i$ is at most one-dimensional, N is spanned by some subset of the basis elements $q_{i,\ell}$. If $q_{i,r} \in N$, then stability under the action of the arrows implies

$$q_{i,r+1}, q_{i,r+2}, \dots, q_{i,c_i-1} \in N.$$

Therefore

$$N = J^s P_i$$

for some s. Hence the submodules of P_i are totally ordered, and P_i is uniserial.

The modules $P_i = Ae_i$ form a complete set of indecomposable projective A-modules. The same argument applied to the opposite algebra shows that the indecomposable injective A-modules are also uniserial. Therefore A is a finite-dimensional Nakayama algebra. □

Lemma 3.4. Let B be a finite-dimensional Nakayama algebra. Then every indecomposable finite-dimensional B-module is uniserial.

Proof. Use the Artinian characterization of Nakayama algebras. For a finite-dimensional algebra B, the following are equivalent:

1. B is Nakayama;
2. every indecomposable finite-dimensional B-module is uniserial;
3. every finite-dimensional B-module is a direct sum of uniserial modules.

Since B is finite-dimensional, it is Artinian, so this characterization applies. □

Definition 3.5. For $i, j \in Q_0$, let

$$\delta(i,j) \in \{0, \dots, L\}$$

be the unique integer such that

$$j \equiv i + \delta(i,j) \pmod{L+1}.$$

Whenever

$$\delta(i,j) + 1 \le c_i,$$

define the interval module $M(i,j)$ by

$$M(i,j)_v = \begin{cases} k, & \text{if } v \equiv i + \ell \pmod{L+1} \text{ for some } 0 \le \ell \le \delta(i,j), \\ 0, & \text{otherwise,} \end{cases}$$

and

$$M(i,j)_{a_m} = \begin{cases} \mathrm{id}_k, & \text{if } m \equiv i + \ell \pmod{L+1} \text{ for some } 1 \le \ell \le \delta(i,j), \\ 0, & \text{otherwise.} \end{cases}$$

The condition $\delta(i,j) + 1 \le c_i$ is exactly the condition that the relations in I are satisfied.

Theorem 3.6. Every nonzero indecomposable finite-dimensional A-module is isomorphic to a unique interval module $M(i,j)$ with

$$\delta(i,j) + 1 \le c_i.$$

Equivalently, the indecomposable A-modules are given by:

$$M(i, i+\ell-1), \qquad 0 \le i \le L, \quad 1 \le \ell \le c_i,$$

where indices are taken modulo $L+1$.

Proof. Let M be a nonzero indecomposable finite-dimensional A-module. Since A is a finite-dimensional Nakayama algebra, the previous lemma implies that M is uniserial. Hence M has a unique maximal submodule, and therefore its top is simple:

$$\operatorname{top} M := M/JM \cong S_i$$

for a unique vertex i.

Let

$$\pi : P_i \twoheadrightarrow M$$

be the projective cover of M. Such a projective cover exists because A is finite-dimensional, and it must be P_i because $\operatorname{top} P_i \cong S_i$. Let $K = \ker \pi$. Then

$$M \cong P_i/K.$$

Since P_i is uniserial, its submodules are precisely

$$P_i,\ JP_i,\ J^2P_i, \dots,\ J^{c_i}P_i = 0.$$

Because $M \neq 0$, the kernel is not all of P_i. Thus there is a unique integer ℓ with $1 \leq \ell \leq c_i$ such that

$$K = J^\ell P_i.$$

Therefore

$$M \cong P_i/J^\ell P_i.$$

As a representation of the quiver, $P_i/J^\ell P_i$ has one-dimensional vector spaces at the vertices

$$i,\ i+1,\ \dots,\ i+\ell-1$$

and zero vector spaces elsewhere. The arrow maps along this interval are id_k, and all other arrow maps are zero. Hence

$$P_i/J^\ell P_i \cong M(i, i+\ell-1).$$

Conversely, for every i and every $1 \leq \ell \leq c_i$, the interval module $M(i, i+\ell-1)$ is isomorphic to $P_i/J^\ell P_i$. Since it is a nonzero quotient of a uniserial module, it is uniserial, hence indecomposable.

Finally, uniqueness follows because the top of $M(i, i+\ell-1)$ is S_i, which determines i, and its composition length is ℓ, which determines the endpoint $i+\ell-1$. This proves the classification. □

Definition 3.7. Let

$$c_0 = L+1, \qquad c_i = L \quad \text{for } 1 \leq i \leq L.$$

Define the set of admissible intervals by

$$\mathcal{I} := \{(i,\ell) \mid i \in \{0, \dots, L\},\ 1 \leq \ell \leq c_i\}.$$

For $(i,\ell) \in \mathcal{I}$, write

$$j(i,\ell) := i+\ell-1 \pmod{L+1}.$$

The corresponding interval module is

$$M_\ell(i) := M(i, j(i,\ell)).$$

Equivalently, $M_\ell(i)$ is supported on the nonzero path

$$i \to i+1 \to \cdots \to i+\ell-1,$$

with one-dimensional vector spaces along this path and identity maps along the arrows of the path.

The number of indecomposables is

$$|\mathcal{I}| = c_0 + \sum_{i=1}^{L} c_i = (L+1) + L^2 = L^2 + L + 1.$$

Corollary 3.8 (Krull–Schmidt decomposition for $\mathrm{Rep}_d(A)$)**.** Every finite-dimensional A-module M admits a decomposition

$$M \cong \bigoplus_{(i,\ell)\in\mathcal{I}} M_\ell(i)^{\oplus m_{i,\ell}},$$

where $m_{i,\ell} \in \mathbb{Z}_{\geq 0}$, and only finitely many multiplicities are nonzero.

Moreover, the multiplicities $m_{i,\ell}$ are uniquely determined by M. Equivalently, the decomposition is unique up to permutation and isomorphism of the indecomposable summands.

Proof. Since A is finite-dimensional, the category of finite-dimensional A-modules is a Krull–Schmidt category. Hence every finite-dimensional A-module decomposes as a finite direct sum of indecomposable A-modules, and this decomposition is unique up to permutation and isomorphism of the indecomposable summands.

By the classification theorem, every nonzero indecomposable A-module is isomorphic to an interval module

$$M_\ell(i) \cong P_i/J^\ell P_i$$

for a unique pair

$$(i,\ell) \in \mathcal{I}.$$

Therefore every finite-dimensional A-module has a decomposition of the form

$$M \cong \bigoplus_{(i,\ell)\in\mathcal{I}} M_\ell(i)^{\oplus m_{i,\ell}}.$$

The uniqueness statement in Krull–Schmidt implies that the multiplicities $m_{i,\ell}$ are uniquely determined. □

3.2 From multiplicity patterns to rank patterns

Definition 3.9 (Multiplicity patterns)**.** Consider an A-module M of dimension vector d and its Krull–Schmidt decomposition into $M_\ell(i)$ for $(i,\ell) \in \mathcal{I}$. For each vertex k, the dimension d_k of the representation M at vertex k satisfies the dimension constraint

$$d_k = \sum_{(i,\ell)\in\mathcal{I}} m_{i,\ell}\, \mathbf{1}\{k \in \{i, i+1, \dots, i+\ell-1\} \pmod{L+1}\}, \qquad k = 0, \dots, L.$$

Define the set of *multiplicity patterns* as

$$\mathcal{M}_d^+ := \left\{ (m_{i,\ell}) \in \mathbb{Z}_{\geq 0}^{\mathcal{I}} \;\middle|\; \begin{array}{l} d_k = \sum_{(i,\ell)\in\mathcal{I}} m_{i,\ell}\, \mathbf{1}\{k \in \{i, i+1, \dots, i+\ell-1\} \pmod{L+1}\}, \\ \text{for } k = 0, \dots, L \end{array} \right\}.$$

We want to stratify the critical loci in terms of the rank patterns, as they are more intuitive and native to the DLN representation. In this section we show that there is a bijection between the multiplicity patterns and a set of cyclic rank patterns satisfying a positivity constraint.

Definition 3.10. For $(a,b) \in \mathcal{I}$, let

$$q_{ab}$$

denote the unique nonzero oriented path of A over the interval $[a,b]$ from vertex a to vertex b.

Definition 3.11 (Cyclic rank pattern)**.** Let M be an A-module of dimension vector d. Its lifted cyclic rank pattern is the array

$$\rho(M) = (\rho_{ab}(M))_{(a,b)\in\mathcal{I}}, \qquad \rho_{ab}(M) := \operatorname{rank}\big(q_{ab}(M)\big).$$

Two coordinates play a distinguished role:

$$\rho_{0L}(M) = \operatorname{rank}(q_{0L}(M)), \qquad \rho_{L,0}(M) = \operatorname{rank}(q_{L,0}(M)).$$

In the DLN representation these are exactly the end-to-end rank and the rank of the closing arrow.

Proposition 3.12 (Forward formula)**.** Let M have multiplicity pattern

$$m = (m_{uv})_{(u,v)\in\mathcal{I}}.$$

Then for every $(a,b) \in \mathcal{I}_L^{\mathrm{wt}}$,

$$\rho_{ab}(M) = \sum_{\substack{(u,v)\in\mathcal{I} \\ u\le a\le b\le v}} m_{uv}.$$

Proof. On an indecomposable interval module $M(u,v)$, the path q_{ab} acts with rank 1 exactly when the path interval $[a,b]$ is contained in $[u,v]$, and otherwise with rank 0. The claim follows by additivity of rank on direct sums. □

Proposition 3.13 (Inversion formula)**.** Extend every rank pattern by the convention

$$\rho_{uv} = 0 \qquad \text{for } (u,v) \notin \mathcal{I}.$$

Then for every admissible interval $(a,b) \in \mathcal{I}$,

$$m_{ab} = \rho_{ab} - \rho_{a-1,b} - \rho_{a,b+1} + \rho_{a-1,b+1}.$$

Proof. Insert the forward formula into the right-hand side and note that, for every fixed interval (u,v), the coefficient of m_{uv} is

$$\mathbf{1}_{u\le a\le b\le v} - \mathbf{1}_{u\le a-1\le b\le v} - \mathbf{1}_{u\le a\le b+1\le v} + \mathbf{1}_{u\le a-1\le b+1\le v}.$$

This coefficient equals 1 precisely when $(u,v) = (a,b)$ and equals 0 otherwise. A more detailed telescoping proof is recorded in Appendix 7.1. □

Definition 3.14 (Admissible rank-pattern set)**.**

$$\mathcal{R}_d := \left\{ \rho \in \mathbb{Z}_{\ge 0}^{\mathcal{I}} \;\middle|\; \begin{array}{r} \rho_{aa} = d_a \text{ for every } 0 \le a \le L, \\ \rho_{ab} - \rho_{a-1,b} - \rho_{a,b+1} + \rho_{a-1,b+1} \ge 0 \\ \text{for every } (a,b) \in \mathcal{I} \end{array} \right\}.$$

Proposition 3.15 (Multiplicity-rank bijection)**.** The multiplicity-pattern set and the rank-pattern set are canonically in bijection:

$$\mathcal{M}_d^+ \xrightarrow{\sim} \mathcal{R}_d.$$

Proof. The forward map sends a multiplicity pattern to the rank pattern given by Proposition 3.12. The inverse map is the finite-difference formula of Proposition 3.13. The diagonal condition $\rho_{aa} = d_a$ ensures that the resulting multiplicity pattern has dimension vector d. □

4 Stratification of the critical loci with cyclic rank patterns

We have identified the cyclic variety Γ_d with the representation variety of the Nakayama algebra $A = kQ/I$ with dimension vector d. Under this identification, the G_d-orbits in Γ_d are precisely the isomorphism classes of A-modules with dimension vector d. By the Krull–Schmidt theorem, each such module has a unique decomposition into indecomposable Nakayama modules, and hence each orbit is indexed by a multiplicity pattern $m \in \mathcal{M}_d^+$.

Proposition 4.1 (Orbit stratification of Γ_d by indecomposable Nakayama modules)**.** Fix $L \geq 1$, let $Q = C_{L+1}$ be the oriented cyclic quiver on vertices $\{0, \ldots, L\}$, and let $A = kQ/I$ be the Nakayama quiver algebra defined above. Then

$$\Gamma_d = \bigsqcup_{m \in \mathcal{M}_d^+} \mathcal{O}_m = \bigsqcup_{m \in \mathcal{M}_d^+} G_d \cdot \phi_m,$$

where ϕ_m is any representative whose associated A-module has multiplicity pattern m. In particular, the G_d-orbits in Γ_d are parameterized by the multiplicity data of indecomposable A-modules.

Using the multiplicity-rank bijection of Proposition 3.15, we may equivalently index the same orbits by cyclic rank patterns. Thus we write

$$\Gamma_d = \bigsqcup_{\rho \in \mathcal{R}_d} \mathcal{O}_\rho,$$

where $\mathcal{O}_\rho$ is the G_d-orbit with cyclic rank pattern ρ. We use the coordinatewise rank order

$$\rho' \leq_{\mathrm{rk}} \rho \quad \Longleftrightarrow \quad \rho'_{ab} \leq \rho_{ab} \quad \text{for every cyclic rank coordinate } (a,b) \in \mathcal{I}.$$

The orbit stratification of the cyclic variety satisfies the closure formula

$$\overline{\mathcal{O}_\rho}^{\Gamma_d} = \bigsqcup_{\substack{\rho' \in \mathcal{R}_d \\ \rho' \leq_{\mathrm{rk}} \rho}} \mathcal{O}_{\rho'}.$$

Equivalently, the frontier of $\mathcal{O}_\rho$ in Γ_d is the union of the strictly lower coordinatewise rank-pattern orbits.

We now specialize this stratification to a fixed S-critical locus. Put

$$W_S := P_S W^\star, \qquad W_S^\dagger := \Sigma_{XY} P_Q,$$

and recall that

$$\Gamma_d^S := \pi^{-1}(W_S, W_S^\dagger) \subseteq \Gamma_d.$$

By Proposition 2.1, the map

$$\Theta_S : \mathcal{C}_d^S \longrightarrow \Gamma_d^S, \qquad \Theta_S(\theta) = (\theta, W_S^\dagger),$$

identifies the S-critical locus with the fiber Γ_d^S.

Let

$$r := \mathrm{rank}(W_S), \qquad q_S := \mathrm{rank}(W_S^\dagger).$$

Under the nondegeneracy assumptions of the main theorem, $r = |S|$ and $q_S = d_L - r$. The two distinguished cyclic rank coordinates satisfy

$$\rho_{0L} = \mathrm{rank}(\mu(\theta)), \qquad \rho_{L0} = \mathrm{rank}(W_0).$$

Hence every point of Γ_d^S has

$$\rho_{0L} = r, \qquad \rho_{L0} = q_S.$$

These rank conditions are necessary but not sufficient for a cyclic orbit to meet the fixed fiber Γ_d^S.

For each $\rho \in \mathcal{R}_d$, define the induced fixed-S stratum

$$\mathcal{O}_\rho^S := \mathcal{O}_\rho \cap \Gamma_d^S = \mathcal{O}_\rho \cap \pi^{-1}(W_S, W_S^\dagger).$$

We keep only the nonempty intersections and define

$$\mathcal{R}_d^S := \left\{ \rho \in \mathcal{R}_d \;\middle|\; \rho_{0L} = r,\ \rho_{L0} = q_S,\ \mathcal{O}_\rho \cap \Gamma_d^S \neq \varnothing \right\}.$$

Thus

$$\Gamma_d^S = \bigsqcup_{\rho \in \mathcal{R}_d^S} \mathcal{O}_\rho^S.$$

The relevant symmetry group for the fixed fiber is the stabilizer of the pair $(W_S, W_S^\dagger)$. Define

$$K_S := \left\{ g \in G_d \;\middle|\; g_L W_S g_0^{-1} = W_S,\ g_0 W_S^\dagger g_L^{-1} = W_S^\dagger \right\}.$$

The equivariance of π,

$$\pi(g \cdot (\theta, W_0)) = (g_L \mu(\theta) g_0^{-1},\ g_0 W_0 g_L^{-1}),$$

shows that K_S acts on Γ_d^S.

Theorem 4.2 (Induced orbit stratification of the S-critical locus)**.** For every $\rho \in \mathcal{R}_d^S$, the stratum

$$\mathcal{O}_\rho^S = \mathcal{O}_\rho \cap \Gamma_d^S$$

is a single K_S-orbit. Consequently, by Proposition 2.2,

$$\Gamma_d^S = \bigsqcup_{\rho \in \mathcal{R}_d^S} \mathcal{O}_\rho^S$$

is a finite stratification into smooth locally closed K_S-orbits. Define the fixed-S closure relation by

$$\rho' \leq_S \rho \quad \Longleftrightarrow \quad \mathcal{O}_{\rho'}^S \subseteq \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S}.$$

Equivalently, since the strata are K_S-orbits,

$$\rho' \leq_S \rho \quad \Longleftrightarrow \quad \mathcal{O}_{\rho'}^S \cap \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \neq \varnothing.$$

Then

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} = \bigsqcup_{\substack{\rho' \in \mathcal{R}_d^S \\ \rho' \leq_S \rho}} \mathcal{O}_{\rho'}^S.$$

Equivalently, if

$$\rho' <_S \rho \quad \Longleftrightarrow \quad \rho' \leq_S \rho \text{ and } \rho' \neq \rho,$$

then the frontier satisfies

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \setminus \mathcal{O}_\rho^S = \bigsqcup_{\substack{\rho' \in \mathcal{R}_d^S \\ \rho' <_S \rho}} \mathcal{O}_{\rho'}^S.$$

Moreover, the fixed-S closure order refines the ambient coordinatewise rank order:

$$\rho' \leq_S \rho \quad \Longrightarrow \quad \rho' \leq_{\mathrm{rk}} \rho.$$

Thus

$$\{\rho' \in \mathcal{R}_d^S \mid \rho' \leq_S \rho\} \subseteq \{\rho' \in \mathcal{R}_d^S \mid \rho' \leq_{\mathrm{rk}} \rho\}.$$

The inclusion may be strict. Via the identification $\mathcal{C}_d^S \cong \Gamma_d^S$, define

$$\mathcal{C}_\rho^S := \Theta_S^{-1}(\mathcal{O}_\rho^S).$$

Then

$$\mathcal{C}_d^S = \bigsqcup_{\rho \in \mathcal{R}_d^S} \mathcal{C}_\rho^S.$$

Consequently, we have a stratification of the variety of the critical points:

$$\mathcal{C}_d = \bigsqcup_r \mathcal{C}_d^r = \bigsqcup_r \bigsqcup_{|S|=r} \mathcal{C}_d^S = \bigsqcup_r \bigsqcup_{|S|=r} \bigsqcup_{\rho \in \mathcal{R}_d^S} \mathcal{C}_\rho^S.$$

Equivalently, after identifying $\mathcal{C}_d^S$ with Γ_d^S, the last strata may be denoted by the corresponding K_S-orbits $\mathcal{O}_\rho^S$.

Remark 4.3 (Compatibility of the fixed product fiber and choice of base field)**.** We now discuss the base field. The representation-theoretic constructions are valid over both $k = \mathbb{R}$ and $k = \mathbb{C}$. The cyclic variety $\Gamma_{d,k}$, the group $G_{d,k}$, and the orbit strata $\mathcal{O}_{\rho,k}$ are defined over k. The orbit decomposition is field-independent: the indecomposable Nakayama modules have representatives with matrices whose entries lie in $\{0, 1\}$. Thus the same cyclic rank patterns occur over $\mathbb{R}$ and over $\mathbb{C}$, and the k-rational points of each nonempty orbit are Zariski dense in that orbit.

Recall now that the fixed pair $(W_S, W_S^\dagger)$ satisfies

$$W_S W_S^\dagger = 0, \qquad W_S^\dagger W_S = 0.$$

Let

$$r := \operatorname{rank}(W_S), \qquad q_S := \operatorname{rank}(W_S^\dagger).$$

The pair $(W_S, W_S^\dagger)$ is then a representation of the two-vertex radical-square-zero cyclic quiver

$$V_0 \rightleftarrows V_L, \qquad BE = 0, \quad EB = 0.$$

Such pairs are classified, over either $\mathbb{R}$ or $\mathbb{C}$, by the two ranks r and q_S. Equivalently, after choosing bases, the pair is conjugate to the normal form

$$B_{r,q_S}(u, v, w) = (u, 0, 0), \qquad E_{r,q_S}(a, b, c) = (0, b, 0),$$

with respect to decompositions

$$k^{d_0} \cong k^r \oplus k^{q_S} \oplus k^{d_0 - r - q_S}, \qquad k^{d_L} \cong k^r \oplus k^{q_S} \oplus k^{d_L - r - q_S},$$

whenever these dimensions are nonnegative. This normal form is defined over $\{0, 1\}$, so the corresponding real and complex orbits have the same algebraic geometry.

Therefore, for the fixed fiber

$$\Gamma_{d,k}^S := \pi_k^{-1}(W_S, W_S^\dagger),$$

the induced strata

$$\mathcal{O}_{\rho,k}^S := \mathcal{O}_{\rho,k} \cap \Gamma_{d,k}^S$$

are independent of the choice $k = \mathbb{R}$ or $k = \mathbb{C}$, provided all objects are taken over the same base field. More precisely,

$$\mathcal{O}_{\rho,k}^S \neq \varnothing$$

if and only if the outer cyclic rank coordinates satisfy

$$\rho_{0L} = r, \qquad \rho_{L0} = q_S.$$

Equivalently,

$$\mathcal{R}_d^S = \{\rho \in \mathcal{R}_d \mid \rho_{0L} = r,\ \rho_{L0} = q_S\}.$$

If one does not want to assume the compatibility condition explicitly, the safe definition is instead

$$\mathcal{R}_d^S := \{\rho \in \mathcal{R}_d \mid \mathcal{O}_\rho \cap \pi^{-1}(W_S, W_S^\dagger) \neq \varnothing\}.$$

For every nonempty induced stratum, the stabilizer

$$K_S = \left\{ g \in G_d \;\middle|\; g_L W_S g_0^{-1} = W_S,\ g_0 W_S^\dagger g_L^{-1} = W_S^\dagger \right\}$$

acts transitively:

$$\mathcal{O}_\rho^S = K_S \cdot x$$

for any $x \in \mathcal{O}_\rho^S$. Hence the K_S-orbit stratification and its Zariski frontier relation are the same over $\mathbb{R}$ and $\mathbb{C}$, after complexification.

The only distinction we do not identify across base fields is the ordinary real analytic connected-component decomposition. A real locus of a complex algebraic orbit may have more than one Euclidean connected component. This does not affect the cyclic rank-pattern labels, the K_S-orbit stratification, the Zariski closure relation, or the codimension computations.

5 Discussion

We have shown that the critical locus of a deep linear network admits a stratification by geometric invariants: the cyclic rank patterns of the associated cyclic quiver representation. More precisely, in the realizable setting with whitened input, each fixed S-critical locus is identified with the fiber

$$\Gamma_d^S = \pi^{-1}(W_S, W_S^\dagger)$$

inside the cyclic variety, and this fiber decomposes into K_S-orbits

$$\mathcal{O}_\rho^S = \mathcal{O}_\rho \cap \Gamma_d^S$$

indexed by the admissible cyclic rank patterns

$$\rho \in \mathcal{R}_d^S.$$

Thus the critical locus is decomposed into pieces indexed by the data

$$(r, S, \rho),$$

where $r = |S|$, S is a subset of teacher singular directions, and ρ is an admissible cyclic rank pattern for the corresponding fixed fiber. This gives a combinatorial description of the critical points of deep linear networks.

This stratification gives a coarse-grained picture of the loss landscape. One may view the critical locus as a partially ordered graph whose nodes are the strata $\mathcal{O}_\rho^S$, or their corresponding

strata inside $\mathcal{C}_d^S$, and whose edges are possible transitions between strata. For a fixed subset S, the relevant order is the fiberwise closure order

$$\rho' \leq_S \rho \quad \Longleftrightarrow \quad \mathcal{O}_{\rho'}^S \subseteq \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S}.$$

This order is compatible with the ambient coordinatewise cyclic rank order:

$$\rho' \leq_S \rho \quad \Longrightarrow \quad \rho' \leq_{\mathrm{rk}} \rho.$$

Hence, within a fixed S-critical locus, specialization can only decrease cyclic ranks. In this precise sense, higher-rank strata are less degenerate, whereas lower-rank strata lie deeper in the boundary of a stratum. The converse need not hold: not every coordinatewise lower rank pattern appears in a fixed fiber, and even among the nonempty fiber strata the relation $\leq_S$ can be strictly finer than $\leq_{\mathrm{rk}}$.

This extends the quiver-geometric perspective on multiplication fibers studied in Lehalleur and Rimányi [2024]. In the realizable case, the global minima are described by the multiplication fiber

$$\mu^{-1}(M),$$

which is the bottom of the loss landscape. The present construction places this fiber inside a larger family of critical loci, indexed by subsets S of teacher singular directions, and equips each such locus with a compatible cyclic-rank stratification. Thus the same representation-theoretic invariants that describe multiplication fibers also organize non-global critical points, including saddle strata.

The cyclic rank patterns also provide natural observables for studying training dynamics. Unlike the rank of the student or the loss value alone, a cyclic rank pattern records the ranks of all partial products around the associated cyclic quiver. These ranks measure how information is distributed across layers and how degenerate the corresponding factorization is. They therefore provide a discrete coordinate system for discussing possible implicit biases of optimization algorithms such as gradient descent or stochastic gradient descent.

The exact dynamics of deep linear networks at small initialization show that the teacher singular values are learned sequentially Saxe et al. [2019]. From the present geometric viewpoint, this suggests that the saddle-to-saddle dynamics biases trajectories toward critical strata with smaller rank patterns that are more degenerate. A possible future formulation is to associate to each rank-pattern stratum an effective energetic and entropic contribution, leading to a coarse-grained description of SGD in terms of a free-energy-like functional. In such a description, the entropy term would reflect the volume of a stratum with a given cyclic rank pattern, or a continuous analogue obtained from the singular values of the partial products of the network.

5.1 Future work

A first direction is to derive explicit codimension formulas for all rank-pattern strata in the critical locus. The results of Lehalleur and Rimányi [2024] compute the codimension and the top-dimensional irreducible components of the fibers of the multiplication map over global minima. In the present setting, one would like analogous formulas for the strata

$$\mathcal{O}_\rho^S = \mathcal{O}_\rho \cap \pi^{-1}(W_S, W_S^\dagger),$$

including both the global-minimum fiber and the saddle and local minima strata. Such formulas would give a quantitative description of the size and singularity structure of each critical stratum. They may also lead to local invariants, such as local codimensions or local real log-canonical thresholds, for the loss near non-global critical points.

A second direction is to relax the simplifying assumptions used in this paper. We have worked in the realizable whitened setting, where

$$\Sigma_X = I, \qquad Y = MX, \qquad W^\star = M.$$

For non-whitened inputs, the same geometry should persist after replacing the teacher matrix M by the whitened teacher

$$M_{\mathrm{wh}} = M\Sigma_X^{1/2}.$$

It would be useful to state and prove the full non-whitened version explicitly, including the corresponding specialization of the closing arrow

$$W_S^\dagger = \Sigma_{XY} P_Q.$$

One could also relax genericity assumptions on the teacher. When singular values have multiplicities, the indexing by subsets S of individual singular directions should be replaced by a formulation in terms of spectral subspaces.

A third direction is to use cyclic rank patterns to coarse-grain training dynamics. Using the combinatorial description of the DLN loss landscape, one can model near-critical stochastic dynamics by a graph whose nodes are strata and whose edges represent possible degenerations or transitions between nearby geometric regimes. Such a graph model could provide a finite-state approximation of how SGD moves between geometrically distinct regions of the loss landscape. Making this precise would require deriving transition rates or effective potentials from the original stochastic process.

Finally, it would be interesting to extend the representation-theoretic perspective beyond linear networks. ReLU networks, for example, are piecewise-linear and decompose into activation regions. On each activation region, the network behaves like a structured linear map, suggesting that quiver- or representation-theoretic invariants may still be useful locally. A stratification for such models would have to incorporate both activation patterns and matrix-factorization ranks. Developing these invariants could provide a combinatorial description of more general neural network loss landscapes.

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7 Appendix

7.1 Inversion Formula

We give a slightly more explicit version of the proof of Proposition 3.13. For every admissible interval (a, b), the forward formula gives

$$\rho_{ab} = \sum_{u \le a \le b \le v} m_{uv}.$$

Therefore

$$\rho_{ab} - \rho_{a-1,b} - \rho_{a,b+1} + \rho_{a-1,b+1} = \sum_{u,v} m_{uv}\, \Xi_{uv},$$

where

$$\Xi_{uv} := \mathbf{1}_{u \le a \le b \le v} - \mathbf{1}_{u \le a-1 \le b \le v} - \mathbf{1}_{u \le a \le b+1 \le v} + \mathbf{1}_{u \le a-1 \le b+1 \le v}.$$

If $u < a$, then the first two indicators cancel, and likewise the last two indicators cancel. If $v > b$, then the first and third indicators cancel, and likewise the second and fourth indicators cancel. The only way for a nonzero coefficient to remain is therefore

$$u = a, \qquad v = b.$$

In that case

$$\Xi_{ab} = 1.$$

Thus the entire sum reduces to m_{ab}, proving the inversion formula.